



UNIVERSITY OF SANTO TOMAS
COLLEGE OF SCIENCE
Academic Year 2026 - 2027



ADVENTURE WORKS PERFORMANCE ANALYTICS DASHBOARD

WRITTEN REPORT

By McReg Analytics

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I. BUSINESS CONTEXT

Adventure Works Cycles, a fictional company from the free database *AdventureWorks* provided by Microsoft, is determined to continue its success as a manufacturer and distributor of bicycle parts and accessories by selling its products through different channels: directly to primary consumers by Internet Sales and to other businesses by Reseller Sales—encompassing wholesale offers and distribution from its pre-existing network of resellers and distributors.

Despite the company Adventure Works Cycles utilizing two different channels for marketing and distribution—Internet and Reseller Sales—and providing marketing promotions and discounts, its sales performance has only been monitored through static reports, which makes it difficult for the business, marketing, sales, and data strategists and analysts from the company to identify underperforming territories, compare and evaluate the effectiveness of implementing both internet and reseller sales, and assess whether the promotional discounts have been financially and operationally effective for the company in real-time. There is an apparent need for a live dashboard so that Adventure Works Cycles can swiftly identify potential risks and ensure positive growth through making data-driven decisions.

To support Adventure Works Cycles in translating this data into actionable business direction, our data consulting team, McReg Analytics, is determined to explore the data, design a dashboard, and deliver a business intelligence solution that consolidates performance across channels, regions, and product lines into a single, reliable source of truth for the continued success of the company.

II. METHODOLOGY

Adventure Works Cycles implemented the following steps in order to build a comprehensive dashboard in Power BI from the AdventureWorks2025 database.

1. Data Overview and Table Selection

The dataset, AdventureWorksDW2025, consists of transactional data, mainly customer and employee information, product categories, marketing promotions and discounts, reseller networks, sales geographic locations, and consolidated information on reseller and internet sales, along with reseller sales quota.

Once the database has been restored, it has been inspected in order to confirm its structure and to be able to see the available tables within the database. An initial query using `sys.tables` and `sys.partitions` was used to catalog every table alongside its RowCounts to establish a baseline understanding of the database and narrow the scope of this project to focus on tables relevant to the analysis. Not all 31 tables were relevant to the analysis that the group was aiming for because of two variables that the group decided would be the standard on what makes a table usable, which are whether they contained data that could be used to answer sales/demand questions and whether they can connect to the fact tables seamlessly. Based on the evaluation, 13 tables were retained: DimCustomer, DimDate, DimEmployee,



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DimGeography, DimProduct, DimPromotion, DimReseller, DimSalesReason, DimSalesTerritory, FactInternetSales, FactInternetSalesReason, FactResellerSales, and FactSalesQuota.

2. Exploratory Data Analysis (EDA)

- Structure Review - querying INFORMATION_SCHEMA.COLUMNS to confirm column names, data types, and nullability for every field, ensuring an accurate understanding of what each table actually contained before writing any downstream logic against it.
- Duplicate detection - grouping each table by its primary key and filtering for any group with more than one occurrence, to confirm key uniqueness before using these fields in relationships.
- Null Value Assessment - this counts nulls across every meaningful column in each table to identify inconsistencies that could distort KPIs.

3. Data Transformation and Schema Simplification

In this stage, we deliberately merged tables in order to simplify the tables to fit in a star schema within Power BI. This includes DimProduct, DimProductSubcategory, and DimProductCategory merged into a single flat DimProduct table to simplify the relationship diagram and reduce the table hops when using DAX queries. A verification check was also made in order to confirm that every subcategory and category key is integrated properly.

4. Data Modeling in Power BI

Each CSV was imported into Power BI Desktop using Import mode (rather than DirectQuery), appropriate given the dataset's manageable size and the need for full DAX functionality. Relationships were then established manually in Model view to reconstruct the intended star schema. Each fact table was connected to its relevant dimension tables using 1 to many relationships.

5. DAX Queries

This is a dedicated measures table that was created to keep all DAX calculations, keeping the model organized and separating these values from the imported data. Measures were grouped by the business area that it corresponds to like the sales/revenue, customer segmentation, reseller and channel comparison, promotions and discounts, seasonality, and profitability. Aside from this, time-intelligence measures rely on the validated DimDate table to ensure accurate period comparisons.

6. Dashboard designs and Visualizations

The final dashboard output was organized into multiple report pages, each corresponding to one of the analytical focus areas: executive overview, channel performance, product performance, marketing performance, customer insights, and employee evaluation. Consistent slicers (date range, product category, sales territory) were applied across pages to allow interactive, synchronized filtering throughout the report. The dashboard was also designed for mobile layout.

III. KEY FINDINGS

Channel Performance

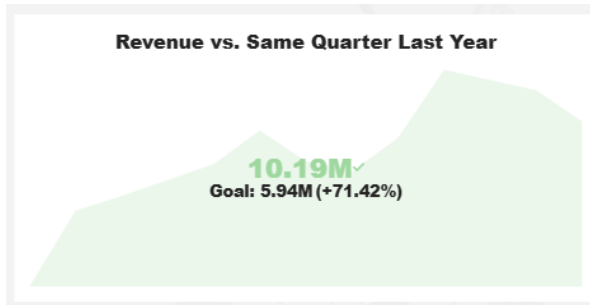


Figure 1. Revenue vs. Same Quarter Last Year

Figure 1 shows the quarter's revenue (Q3 2013) compared against last year's same quarter's revenue (Q3 2012). The findings showed that the business earned +71.42% more revenue this quarter than last quarter. Business growth is evident.



Figure 2. Total Revenue Summary

Figure 3. Total Profit Summary

Figures 2 and 3 show the business's combined revenue and profit and the share of each channel. The findings showed that the largest contributor to revenue was Reseller, with a quarter of revenue from the internet; on the other hand, the largest contributor to profit was Internet, with a small share from Reseller. It can be seen that, even though the reseller is the driving force for revenue, the internet is more valuable for profits.

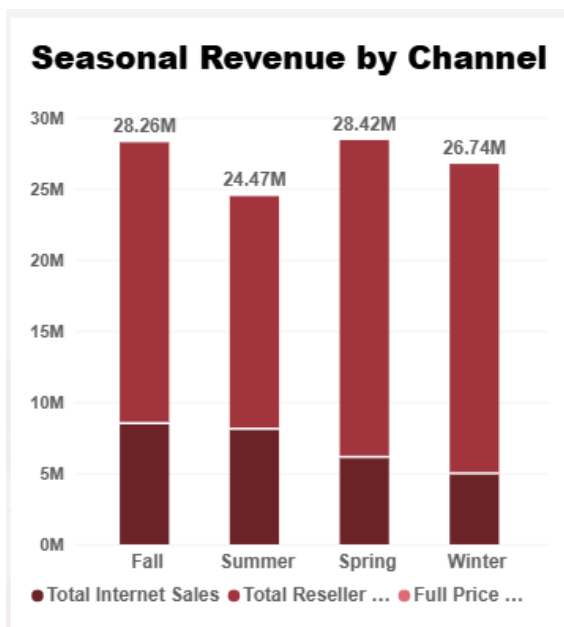


Figure 4. Seasonal Revenue by Channel

Figure 4 shows a stacked column chart, where the two channels' revenue shares can also be seen. Instead of quarters, revenue is analyzed by season to identify seasonal patterns. The seasons are based on the Northern Hemisphere conventions. The findings showed a consistent movement of both channels throughout the seasons. It is also evident that in almost all seasons, revenue is the same, a bit higher in spring, and it just dip a little in summer. This suggests that the product isn't dependent on seasons.

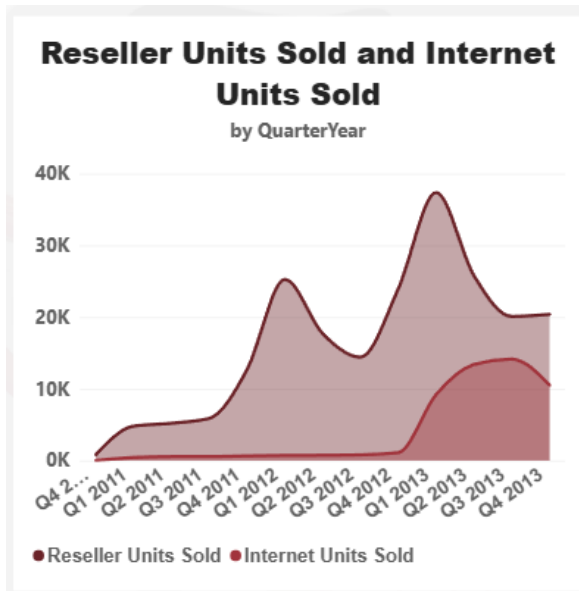


Figure 5. Reseller Units Sold and Internet Units Sold

Figure 5 shows a stacked line chart comparing units across both channels. The findings show not only that resellers sell bikes, but also that they are contributors to units sold. That explains being the driving force in revenue. But a spike in sales under the internet channel can be seen at the start of Q4 2012. This suggests that the internet channel is getting traction in sales. Then, a downward spike is visible on the reseller's channel at the start of Q1 2013. This can show where one channel is climbing while the other is declining. And both can be seen getting close at the end of the quarter.

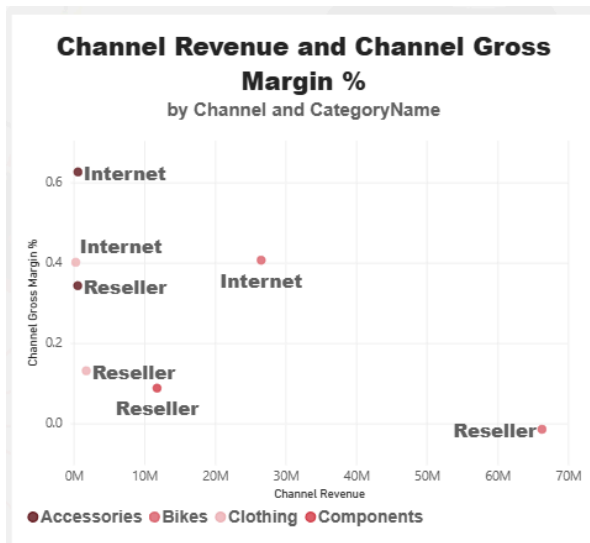


Figure 6. Channel Revenue and Channel Gross Margin %

Figure 6 shows a scatter plot used to analyze the gross margin and revenue for categories that both channels sell. The findings showed that under the reseller channel, bikes have the highest revenue yet the lowest gross margin. While internet channel accessories have among the lowest revenues, they are more profitable.

Product Performance

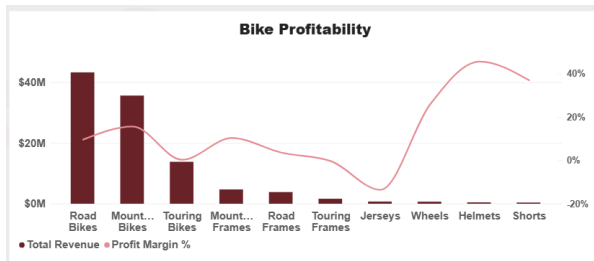


Figure 7. Bike Profitability

Figure 7 shows a combination of a line and a stacked column chart, comparing revenue and profit against the products sold. The findings show that, even though the road bikes lead in revenue, the helmets lead in profits. This further strengthens the earlier findings that the accessories sector is more profitable, even if its revenue isn't notable compared to bikes.



Figure 8. Total Revenue by Promotion Type and Sales Territory Group

Figure 8 shows a stacked column chart that analyzes each territory group and identifies which promotion type drives their market. The findings show that North America accounts for a large share of revenue across all promotions. While Europe shows a consistent revenue across promotions. While Pacific only has a notable revenue share.



Figure 9. Discount Margin Erosion

Figure 9 is a combined line and column chart that shows how much revenue is generated by each promotion or discount. The findings showed that profits are high when there's no discount. A slight profit under the volume discount, while the rest of the profit margin slowly declines to none at all. Yet even when consumers buy much more, which is covered by a volume discount, profit is still not notable, unlike when customers buy without a discount.

Marketing Performance

Promotional Profit Margin % 0.43	Promotional Avg Revenue Per Or... \$975.78
Full Price Profit Margin % 0.41	Full Price Avg Revenue Per Order \$500.87
Reseller Discount Rate 0.7%	Total Discount Given 46.07K

Figure 10. KPI Cards of Rates

Figure 10 is a quick glance at the side-by-side performance status of profit, average revenue, and discount rate. The findings show that the profit margin is 2x more in average revenue per order than at full price. This reflects the volume-discount promotion, not proof that the promotion drives bulk buying or single-product sales.

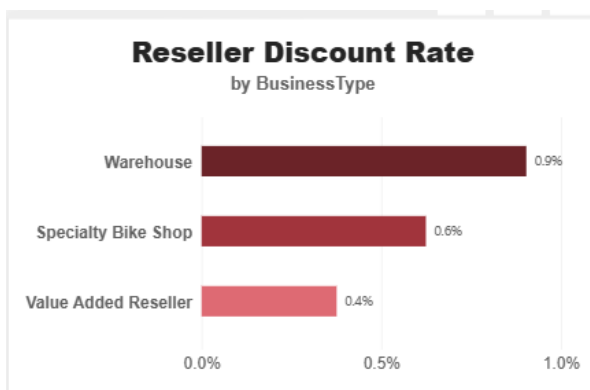


Figure 11. Reseller Discount Rate

This shows a side-by-side bar comparison of reseller types. The findings show that the business gives a slightly higher discount rate to warehouses compared to others. Since warehouses tend to buy many more products, the business prefers to offer them discounts rather than the rest.

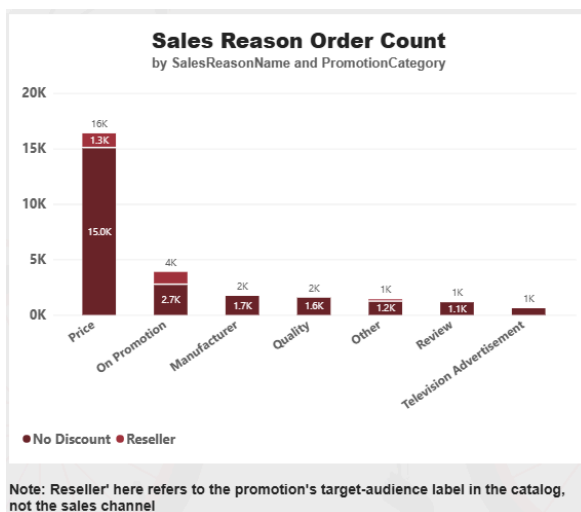


Figure 12. Sales Reason Order Count

Figure 12 is a stacked column chart showing why a customer buys such a product. This is only available from the internet channel. The reseller part there isn't the channel itself. But a promotion that's supposed to be for resellers, since they bulk buy, but some customers also tend to buy in volumes, hence the promotion category labeled them as such. The findings showed that price is a strong reason why customers buy such product/s. This shows that this business offers reasonable, friendly prices to consumers.

Customer Insights

	Occupation	Sum of YearlyIncome	CommuteDistance	Sum of NumberCarsOwned	Sum of NumberChildrenAtHome	Internet Sales
rs	Clerical	20000	10+ Miles	4	0	2.2
	Professional	60000	0-1 Miles	4	4	2.2
	Management	100000	10+ Miles	3	5	2.2
nder	Professional	70000	10+ Miles	3	0	2.2
	Professional	80000	10+ Miles	3	0	2.2
itt	Professional	30000	5-10 Miles	2	0	2.2
ts	Clerical	20000	1-2 Miles	2	0	2.2
	Clerical	20000	1-2 Miles	2	0	2.2
s	Professional	60000	10+ Miles	2	1	2.2
ni	Skilled Manual	20000	5-10 Miles	2	0	2.2
	Professional	60000	1-2 Miles	2	0	2.2
	Professional	30000	5-10 Miles	2	0	2.2

Figure 13. Customer Profiles from Internet Channel

This is a table showcasing the customer profile for internet channels only. The findings show that the customer with the highest purchase also tends to have more than one car and has children. It also shows that they have a high income and are about 10 miles away. This shows that this business also caters to families, especially children.

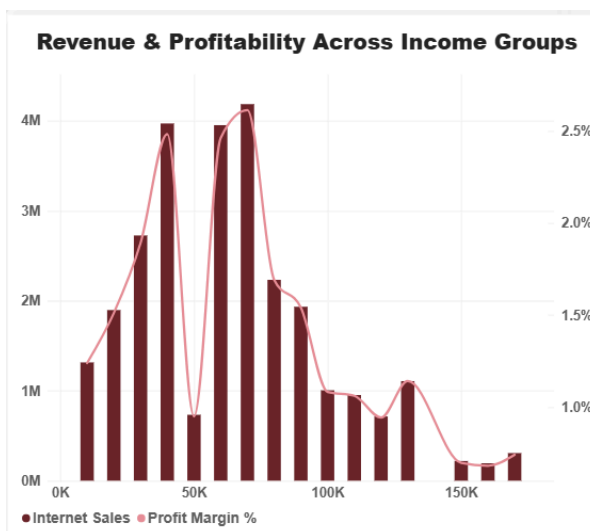


Figure 14. Revenue & Profitability Across Income Groups

Figure 14 is a combined line-and-column chart, showcasing income across groups and business profit for each group. The findings show that middle-income earners have a strong drive for revenue and profits. Compared to the high-income group, where business barely makes revenue and profit from them.

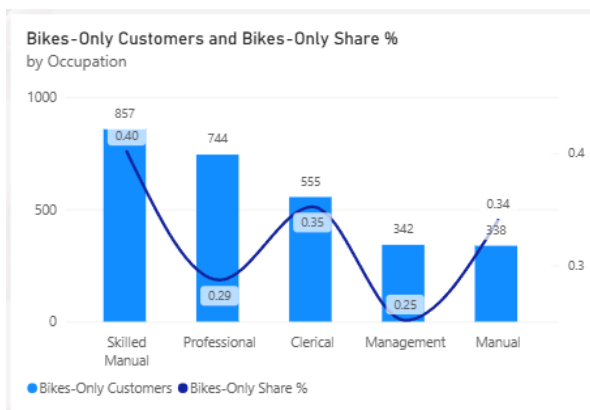


Figure 15. Bikes-Only Customers and Bikes-Only Share %

Figure 15 is a combined line-and-column chart showing customers who buy only bike/s. The Skilled Manual customers are one of the largest group of bikes-only buyers, making them the top target for bundle sales.

Employee Evaluation

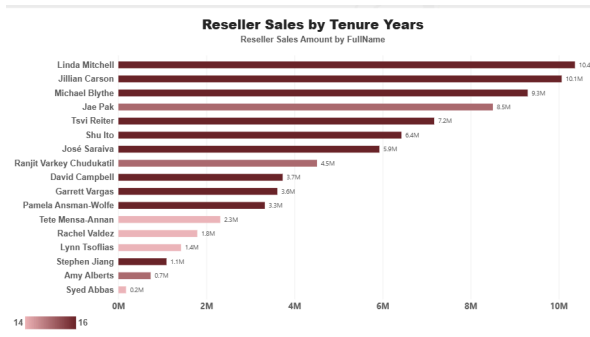


Figure 16. Reseller Sales by Tenure Years

Figure 16 is a bar chart of the sales department, with each employee shown by tenure and sales. The findings show that longer tenure doesn't necessarily mean being among the top sales performers. Some employees with over a decade of experience rank among the top performers, while others with the same tenure fall behind, and some employees with fewer years can keep pace with their colleagues in the top ranks. This shows that individual sales performance isn't dependent on tenure years.

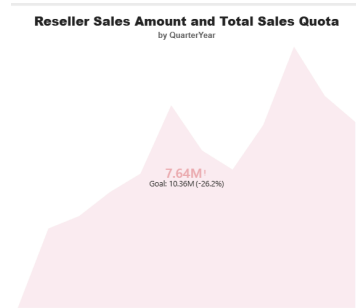


Figure 17. Reseller Sales Amount and Total Sales Revenue

Figure 17 is a KPI visualization comparing total reseller sales to the total sales quota. The findings show that the reseller didn't reach their quota and was down by a large margin. This is evident in the revenue trend over time, which starts to dip in Q4 2013.

IV. ACTION PLAN

During the testing stages of the data, many discrepancies were discovered, but were resolved prior to taking any further steps towards visualizations. To avoid further incidents, it is best to take the following precautionary measures:

Due to the presence of data disparities, such as quality issues, inconsistency, and unsynchronized data reports, it is necessary to use test data across all stages of the value chain before sending actual data through the pipeline to avoid corruption. Furthermore, although the provided dataset did not disclose whether the data records were automated or manually entered, it is best to note that automated data synchronization would be the best option. Manual updates are susceptible to human errors and are not sustainable. If the company insists on manual updates, providing data entry standards can help avoid causing inconsistent data formats.

Additionally, null values remained in the dataset despite Exploratory Data Analysis (EDA). Null values in the dataset are hard to avoid if the business does not have the necessary precautions in place to handle them. If the column containing NULL values can be visibly correlated with other values, it could simply be renamed; or, if the values are important but do not belong in any existing column, it could be moved to a dummy column. Other measures can be used, such as regression models, wherein predictive models are used to assign missing values based on other available data. Interpolation could also be an option, as it fills in missing data by following certain trends and adjacent data points.



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To assess effectiveness, KPIs are highly encouraged. In this case, the following question would be most effective: Has there been a decrease in discrepancies since the implementation of an automated system and the establishment of a data-entry standard?

Similarly, there was no net pricing present in the data. It is easy to be overwhelmed by the volume of recorded data; the best move is to combine similar or related data into a single schema to ensure that all necessary data will be included in future calculations and analyses.

After further analysis, physical data fragmentation is also an issue. Utilizing data warehouses, which store all recorded data in one place, not only improves organization but also guides the business in making smarter decisions by enabling easier analysis and clear guidance on which path to take. If the business is uninterested in utilizing data warehouses, using cloud computing services, a widely used, vast network of online servers, is also an option. It provides flexibility and advanced data management tools in addressing data fragmentation issues. Furthermore, implementing a system where data quality is strictly monitored, not only for accuracy, but also for consistency, given that any gaps in the data can greatly affect the business. Incorporating machine learning tools is also encouraged to automate checking data for further conflicts.

The best questions to ask to ensure effectiveness would be: Are there any more gaps in data since the implementation of any automated system, the use of cloud services, or data warehousing? Did decisions that were based on the data create a positive impact on the business statistics?

McReg Analytics' team also noticed that neither refunds nor returns are recorded, which makes the data slightly unreliable. The simplest solution would be to update the schema to include product returns and refunds to ensure accurate net revenue results. In line with this, the team also noticed imbalances between sales channels. One way to resolve this issue is to use unified channel metrics; they ensure the schema is perfectly symmetrical, meaning all columns present in the resellers' channel are also present in the online channel. Additionally, the schema must be updated to also track promotions made, in order to establish a general target audience for the business. For example, since bike accessories yield a higher profit (Figure 7) than bikes (Figure 6), curating a bundle of accessories that include helmets, knee pads, bicycle lights, and bells, for parents buying a bike for their children, based on the top 10 consumers.

To ensure effectiveness, results must answer these questions: Is there a noticeable improvement in orders and promotional items since the implementation of the improved schema? Did the schema successfully capture any blind spots in the dataset? (e.g., refunds and returns)

After procuring insights, McReg Analytics' team are suggesting the following action steps:

First, as shown in Figure 2 and 3, Internet channel generates 96% of total profits despite driving only 25% of the overall revenue. It would be highly beneficial for Adventure Works to reallocate a portion of its marketing budget to targeted digital and social media advertisements for middle-income Internet customers (Figure 14).

Second, in Figure 2, resellers take in the most money but have extremely low profit in Figure 3 due to slim profit margins (Figure 6), which leads to huge discounts for customers who buy directly from the warehouse (Figure 11). To solve this issue, Adventure Works don't have to remove or reduce any promos in the business; rather, it is best to cap the discounts given. For example, trigger a higher minimum purchase volume for the bulk discount and a maximum 10% volume discount per reseller.

Third, as shown in Figure 9, promotional discounts do not drive any substantial incremental profit, as profits are still at its peak when products are sold with No Discount. As observed, price is the reason why internet customers buy; hence, it is more beneficial to allocate the promotions into



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high-margin bundles. For example, buy 1 Touring-1000 Bike and get taillights at 20% off to drive higher cart value.

Fourth, North America drives the highest revenue, followed by Europe, while the rest of the territories are underperforming in terms of customer activity when responding to discounts as shown in Figure 9. It is suggested to reallocate most of the promotional marketing of high-margin bundles (as previously suggested) into North America and Europe to drive the highest customer responsiveness while still capping high-discount promotions in these core territories. It would also be beneficial to restrict inventory to high-margin lines and halt discounts in underperforming territories.

Fifth, as shown in Figure 7, while Road Bikes generate the highest revenue, these only provide moderate profit margins. Rather than focusing on the production of low-margin lines (e.g. bike models), reallocate production and inventory into margin lines such as Accessories and Apparel (e.g. helmets, shorts, wheels) as these sectors lead to profit margins.

Lastly, the 2013 reseller sales quota was missed by \$7.64M (Figure 17), which could cause a significant loss for the business. In Figure 16, long-term sales employees are outperformed by newer employees. Setting realistic goals and standards for each region can help sales representatives reach quotas.

V. LIMITATIONS AND ASSUMPTIONS

Limitations:

1. Dataset

The dataset used for this analysis presents several limitations, such as Promotion tracking being available only for the Internet channel and the inability to assess its effectiveness for the Reseller channel. Several fields required data correction, including a literal "NULL" in the Product class field, an "NA" in the sales Territory Group, and an inconsistent data format in the Product dimensions end-of-sale field. The Internet and Reseller didn't share a synchronized start and end of report, hence channel comparisons were adjusted to ensure fairness. Sales quota data are available only for the Reseller channel, thereby limiting performance analysis across both channels. Customer demographics were also limited to Internet Sales customers. The dataset doesn't include returns or refunds reports, which could add value to the analysis. And in further analysis, we've noticed that, despite the business opening in 2011, the dataset only has data recorded through 2014, with consistent 2-month gaps for 7 months. It is unclear whether the business was closed in 2014, whether data recording was simply halted, or whether the remaining data for the years since then is stored elsewhere. To conclude, we were only able to narrow the scope to Q4 2011 to Q4 2014, thereby limiting the analysis to more meaningful findings, such as historical patterns.

2. Dashboard

The dashboard ensures fair comparison from both channels, hence setting the analysis window from Q1 2012 to Q3 2013 is the span during which both are available. The Seasonal definitions used Northern Hemisphere conventions; therefore, they don't reflect other countries like Australia, where seasons are reversed. There is also the row-level security, which was not implemented, and the dashboard was not published due to Microsoft licensing constraints. Hence, it is instead delivered as a downloadable file for local viewing. Lastly, since the dataset doesn't include official company-set sales targets for most metrics, quarter-over-quarter and year-over-year comparisons were used in place of fixed KPI benchmarks; performance is based against its own past rather than a set goal.

Assumptions:



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1. The team has assumed that when it comes to the Total Revenue of Reseller and Internet Sales, Reseller has the upper hand because people would want to buy bikes in person since they would want to inspect the product first before buying it.
2. The team has assumed that since the profit margin for Resellers is lower than the Internet Sales is because of the costs of manning a physical store such as wages, taxes, and location.
3. The team has assumed that the highest customers that own multiple cars but also have children, purchased the bikes for their kids rather than personal use.

AI Assistance Usage:

Generative AI tools (specifically Claude and Gemini) were used to assist in coding and filling out the Business Definition of some attributes for the AdventureWorksDW2025 Data Dictionary due to the large number of attributes within the entire dataset. AI assistance was similarly used to help draft and troubleshoot DAX formulas for calculated measures and columns. However, McReg Analytics guarantees that the final content was reviewed, verified, and edited independently by our team, and we take full responsibility for what is placed within the Data Import, Data Modeling, and Technical Documentation.